

Direction Finding and Signal Characterization of Unknown Emitters

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Abstract: Under guidance of the NATO HQ C3 Staff and NATO Capability Panel 2 (CaP2) on Navigation and Identification, an experimental prototype of a GPS jammer direction finding and signal classification system was constructed using Software Defined Radio technology. The system is based on phased array processing of signals from a 6-element uniform linear antenna array, sampled with a multi-channel phase coherent Radio Frequency receiver. The prototype system was tested against realistic jammers and spoofers and demonstrated to have an accuracy of 2 degrees and a resolution around 10 degrees using the MUSIC algorithm. Signal classification features were proven to be very useful tools to support the operator in assessment of the jamming and spoofing situations. This report describes the design, implementation and extensive test results.

1. Introduction

The Global Positioning System (GPS) has become the main source of Position, Navigation and Time (PNT) information, both for military and civilian use. However, with the increase of reliance on GPS came a growing concern about the susceptibility to radio frequency jamming. The signal level received from the distant satellites is extremely weak and can be overpowered or disrupted by malicious transmitters. While jamming mitigation techniques have been fairly successful, it is still important in an operational military environment to monitor the radio frequency spectrum for hostile GPS jamming and spoofing activity. In case any jamming or spoofing is detected, the signal must be analyzed and its source located if possible. The monitoring task described above is typically performed using expensive military grade receivers, usually built and integrated into a system for this specific purpose, but recent developments in Software Defined Radio (SDR) technology look promising. The NATO HQ C3 Staff, in coordination with the NATO C3 Board on Navigation and Identification, tasked the NATO Communication and Information Agency (NCI Agency) to construct an experimental prototype system to detect, classify and localize GPS jammers and spoofers using relatively low-cost SDR devices. The system, named GANDALF (Ground-based Asset designed and built by NCI Agency for Direction And Location Finding), utilizes phased array processing of signals collected with a multichannel phase coherent receive system. During the past two years, the prototype system was subjected to extensive field trials, results of which are provided in this report. The conclusion is that the prototype system provides an accuracy of roughly 2° and a resolution around 10°. Signal classification (signal intelligence) features are demonstrated to provide significant added value: in particular, the spoofing characterization capability provides useful insights on the nature of the spoofing attack (e.g. synchronous, asynchronous or meaconing).

2. System description

Figure 1 shows the GANDALF instrument block diagram. The instrument comprises: antenna subsystem (composed of a phased-array of 6 L1 dipoles, a phased-array of 6 L2 dipoles, two SMA GPS L1 antennas and a hyper log monitoring antenna), switch matrix, 4 transceiver units (each containing 2 independent transceiver channels in one enclosure), a 8-way RF splitter, a network switch and a high speed desktop computer running LabView and Matlab. The antenna and associated electronics are mounted on the top of a transportable telescopic mast. The antenna array frame is constructed from thick-walled plastic tubes. The individual antenna elements are clamped on the frame using brackets, designed with a CAD tool and printed in plastic on a 3D printer. The brackets are easy to dis-assemble for transport and provide the required flexibility in configuration of the array. Using a plotter, water-proof stickers are made to provide a clear scale on the frame so a desired antenna layout can be made precisely and reproduced. A sleeve on the bottom of each antenna element ensures the connector is protected from rain. The antenna system can be configured in different ways depending on the needs. In most of the trials GANDALF has been deployed with two (L1 and L2) Uniform Linear Arrays of 6 dipoles each. A Hyper log monitoring antenna is used to point towards the direction of arrival of the detected spoofer or jammer.

Figure 2 shows the antenna layout along with the deployed configuration of the two arrays and the log monitoring antenna on a transportable mast.

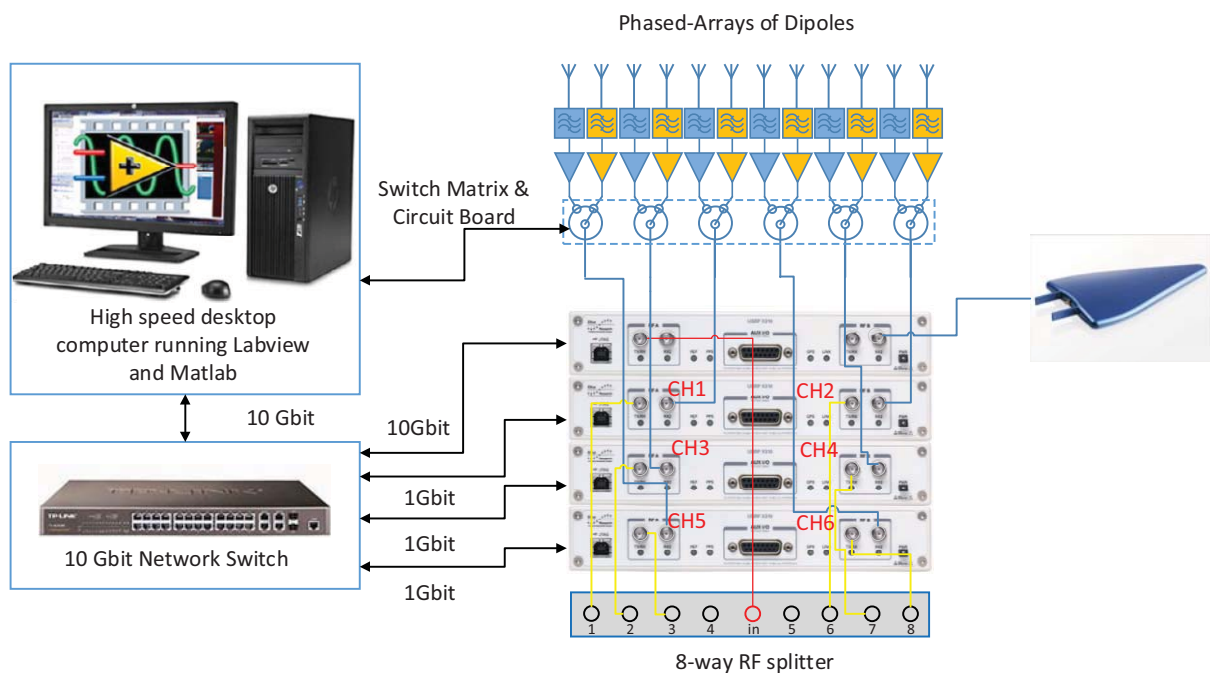


Figure 1. GANDALF instrument block diagram

The dipole's design is a copy of a typical Wi-Fi antenna. The braid of a coaxial cable is separated from the center conductor and "folded" down. A small metal tube is crimped around the braid to form the lower leg of the dipole antenna while the center conductor forms the upper leg. A plastic cover and sleeve provide protection against rain. We have selected 10 cm spacing for L1 and 12 cm spacing for the L2 array antenna. Such distances were proved to be sufficiently small to avoid (nearly all) DOA (Direction Of Arrival) ambiguities.



Figure 2. On the left, layout of the GANDALF arrays of dipoles; on the right, view of the antenna subsystem on a transportable telescopic mast.

Two waterproof GPS antennas by Navilock with SMA connection are used for the automatic true-north alignment of the whole antenna. The distance between the SMA GPS antennas is 80 cm: this spacing was proved to provide a boresight accuracy better 0.5° , by processing GPS L1 signals and performing a carrier phase measurement of the two antennas.

The directional antenna hyper log 7060 by Aaronia is used to point toward the direction of arrival of the jamming or spoofing signals, thus allowing an in-depth investigation and characterisation of the signals source. A switching network is used to connect the mast-mounted antenna arrays to the receive system. The antenna feed, including electronics, is mounted in the top of the mast inside a waterproof enclosure.

The signals from the antenna arrays are routed down to the receivers via a bundle of 6 Radio frequency cables of approximately 12 m length. Because low-loss cables (such as RG-213) are too heavy and stiff when bundled, standard RG-58 cables are used, instead. The downside of these cables is the loss over a 12 m cable run, easily exceeding 10 dB at GPS L1 frequencies. This loss is compensated by amplifying the signals as close to the antenna as practically possible. For the dipoles, ZX60-P162LN+ amplifiers are used: they provide reasonable gain (~ 20 dB) and matching, with low noise figure (~ 1 dB). The tolerance on the supply voltage level to the amplifiers is small (± 0.2 V). For this reason an excess voltage is fed up to the top of the mast and regulated there by means of a standard 74-series regulator IC.

Although the dipoles are tuned to either L1 or L2 frequencies, they still capture significant signal at other frequencies. To avoid this, six L1 bandpass filters VBF-1575+ and six L2 bandpass filters ZX75BP-1100-S+, with 25 dB stop-band rejection, are used before the signals coming from the dipoles are amplified. The SMA GPS antenna elements, instead, do not require additional L1 filtering nor amplification stage as they already have them integrated.

Six ZFBT-4R2G-FT+ Mini-Circuits bias TEE with SMA connectors are used to select up to six antenna elements among the 12 available. The bias TEEs are micro-processor controlled. These components have a wide band of operation, ranging between 10 and 4200 MHz, low insertion loss (0.6 dB typ.) and feed through terminal per DC port.

The receive system is based on the National Instruments Universal Software Radio Peripheral, the NI-USRP 2952. This is a software defined radio reconfigurable device capable of receiving signals in a reasonable large band, ranging between 50 MHz and 2.2 GHz. The USRP-2952 provides an integrated hardware and software solution for rapidly prototyping high-performance Direction Finder. The USRP-2952 is equipped with GPS-disciplined 10 MHz

oven-controlled crystal oscillator (OCXO) reference clock. GPS disciplining delivers improved frequency accuracy and synchronisation capabilities. The selected model contains 2 independent transceiver channels in one enclosure. To build a 6-channel receive system, 3 units are used. The transmit capability of these 3 units is not used.

As explained in following sections, one additional unit is used to provide a reference signal for both amplitude and phase calibration, thus allowing fully phase coherent measurements. On this unit, not only the transmit capability is used but also the receive capability: the hyper log monitoring antenna is, in fact, connected to it. Long recordings of jamming and spoofing events are made by this unit.

A high-end desktop computer running National Instruments LabVIEW and Matlab is used for control, processing and display. To support the capture of sampled RF data at high rates from 6 simultaneous channels, a 10 Giga-bit (Gb) Ethernet card is installed in this computer.

A network switch with two 10 Gb and multiple 1 Gb Ethernet ports is used to connect the computer and all transceivers. In this application the computer is assigned a 10 Gb port because it needs to communicate with all transceivers simultaneously. The transceiver used to transmit the calibration signals (on top of the stack in Figure 1) is assigned with the second 10 Gb port as it is empirically proven to be very sensitive to buffer underruns.

Front-end components set the stage for receiver processing and direction finder. Properly integrated in the USRP-2952 these components enhance the desired signal and mitigate interference and noise, helping to achieve the desired receiver performance.

The GANDALF system is installed in a 4x4 vehicle, to allow for quick deployment in difficult or steep terrain (Figure 3). The antenna is installed on a telescopic antenna mast, with a maximum height of roughly 7 meters. A small 1kW gasoline generator, combined with a UPS can be used to power all equipment, if no external power source is available. The system is operated by two persons.



Figure 3. GANDALF vehicle installation

3. System calibration

3.1. Phase Coherence Approach

In a fully phase coherent multi-channel receive system, the phase difference between channels in the sampled digital data is directly related to the phase difference of the RF signals at the input ports. This relation should be identical even when the system is power cycled or re-tuned. This requires a high level of synchronization between the channels. Although the NI-USRP transceivers are advertised as coherent and equipped with hard- and software features to provide

synchronization, they are not fully phase coherent. After restarting or retuning the transceivers, a random phase difference exists between all 6 channels. This is most likely because each channel derives its own local oscillator from the (shared) 10 MHz reference. The relative phase jumps between the channels are unpredictable and have to be determined every time after (re)tuning. In particular, after starting a new receive session (as is the case when retuning) a phase drift exists between the 3 individual transceivers. Two channels in the same transceiver (i.e. in the same “box”) do not experience this drift.

To overcome this problem, the drifts are measured and compensated for, by briefly disconnecting the signals to be measured in the 3 transceivers and replacing them with a calibration signal of known phase. This is achieved via a programmable RF switch. This method provides very clean calibration and the possibility to use a wideband calibration signal at the cost of short receive interruptions (order of 10 ms).

3.2. Antenna automatic-alignment

Signals from known directions of arrival are used to estimate the azimuth angle of an antenna array. In fact, it is the inverse operation of the direction finding process. The GPS almanac contains a prediction of the orbit parameters for the GPS satellites. A recent almanac allows to calculate the satellite constellation, including elevation, azimuth and Doppler frequency. Almanac data is not accurate enough for calculating the actual navigation solution, but accurate enough to serve as “known direction of arrival” for the purpose of automatic alignment. The auto-alignment procedure is based on carrier phase measurements of two SMA GPS L1 antennas. Using the Almanac and knowing the time and the coordinates (geodetic latitude, longitude and altitude) of the antenna site, it is possible to find the satellites in visibility and compute their elevation and azimuth angles, as well as their distances and Doppler frequencies. This information is used to generate a 10 milliseconds L1 C/A code reference signal for each visible satellite, with whom the GPS data recorded by the two receiver channels is correlated. Only the signals having Signal-to-Noise Ratios (SNR) higher than 40 dB-Hz are retained for processing.

Typically, it is possible to get alignment accuracies better than 0.5° , by processing 10 minutes of GPS data. This procedure reduces the difficulties and disadvantages of the manual alignment by providing a simple method of aligning the antenna array using GPS signals. The alignment method can be performed in conditions of poor visibility, such as at night or in fog. The method is inexpensive and simple to use and provides the user a reliable and accurate way to align the antenna with a predetermine azimuth direction.

4. Phased array processing and direction finding

The prototype system makes use of phased array principles to determine the direction of arrival of signals. Many good text books [1]-[3] and scientific articles have been written on the topic of phased array processing. This section only provides a brief explanation of the two Direction Finding algorithms we have implemented.

The first, and simplest, Direction of Arrival estimation technique implemented for GANDALF is electronic scanning of a beam across the azimuth field of view and finding the maximum received power. This works best in the simple case where only one direction of arrival is present. The algorithm fails in scenarios with more than one signal direction, in particular if closely spaced or with significant difference in signal strength.

The second method implemented for this project is the Multiple Signal Classification (MUSIC) algorithm [1]-[3]. It is a subspace based super-resolution technique and it is suitable in the case of multiple simultaneous signals, particularly when closely spaced. The MUSIC algorithm

provides a method to identify the noise subspace using eigen-analysis. In practice, MUSIC estimates the frequency content of a signal or autocorrelation matrix using an eigenspace method.

5. System operation

The operating modes of the GANDALF system are:

- Normal Mode;
- Direction Finding Mode;
- Emitter Characterization Mode.

Normal Mode periodically searches for either jamming or spoofing events. In practice, GANDALF records snapshots of 20 ms every 3-5 seconds, from the SMA GPS antennas and the dipole antennas. A Matlab software processes in real time the snapshots and search for either jamming or spoofing events by: spectrum analysis, histogram analysis, Jamming-to-Noise (J/N) measurements and Carrier-to-Noise Power Density (C/N0) measurements.

The Direction Finding Mode estimates DOAs and visualizes them on a map. In this mode the signals received by the arrays of dipoles are coherently processed and, depending on the findings, either the beam scanning method or the MUSIC algorithm is used for DOA estimation. The Emitter Characterization Mode retrieves jammer's or spoofer's information by analyzing long recordings of the emitted signal. For GPS spoofing, this mode provides useful information on the spoofing attack, such as: number of spoofed signals, type of attack – asynchronous, synchronous, or meaconing – and intended (virtual) receiver path.

6. Trial results

6.1. Static Jamming Scenarios

Two fixed jammers were positioned across the valley in clear line of sight from GANDALF:

- one “silver” jammer at 420 m distance at an azimuth of approximately 218°;
- one “yellow” jammer at 484 m distance at an azimuth of approximately 255°;

The angular separation between the two jammers was approximately 37°.

For resolution testing, the yellow jammer was moved towards the silver jammer. The jammers produced various waveforms, such as a Continuous Wave (CW) or Spread Spectrum (SS) signals. The center frequencies were programmable.

The following test scenarios were tested:

- 4 single jammer Broad-Band (BB) scenarios.
- 4 single jammer CW scenarios.
- Several scenarios with both jammers simultaneously on.

Figure 4 shows an example of detailed information provided by the GANDALF's software for two single-jammer CW scenarios (1 scenario where only the “silver” jammer transmits and 1 scenario where only the “yellow” jammer transmits). The top left graph shows the 10 kHz section of the spectrum that was used for direction finding processing (beam scanning method). Although both jammers were programmed identical, a 1 kHz difference in frequency was found. The eigenvalue spread, in Figure 4, indicates there is only one dominant eigenvalue, confirming the presence of only one arrival direction per scenario. The largest eigenvalue exceeds the second largest by almost 50dB. The graph (bottom figure) is normalized such that the theoretical best match would produces a value equal to 1. In practice this value is never achieved, due to (among others) a slight multipath, calibration error, component drift or plain

measurement noise. Nevertheless, the peak values were found at the correct directions of arrival, with around 1° RMS error (not shown here).

6.2. Moving Jamming Scenario

A man-portable BB L1 jammer, transmitting a few milliwatts power, was moved along the red path in Figure 5. The actual positions of the jammer and GANDALF were provided by GPS receivers. The distance between GANDALF and the jammer ranged between 500 m and 1 km. The jammer transmission was not continuous as it was switched on and off by manually pressing and releasing a button. As depicted in Figure 5, the GANDALF system provided line of bearings that followed the path of the jammer. Figure 6 (left) shows the measured DOAs as provided by GANDALF versus the expected DOAs, as retrieved from GPS measurements. Figure 6 (right) provides the resulting DOA errors (e.g. the difference between expected and measured DOAs). The error has a mean value of -0.9° and a standard deviation of about 1.8° . These results do not surprise as the error is very much dependent on the measurement environment: multi-path, terrain masking, body masking and degradation of GPS performance by L1 jamming are definitely key factors affecting the measurement accuracy.

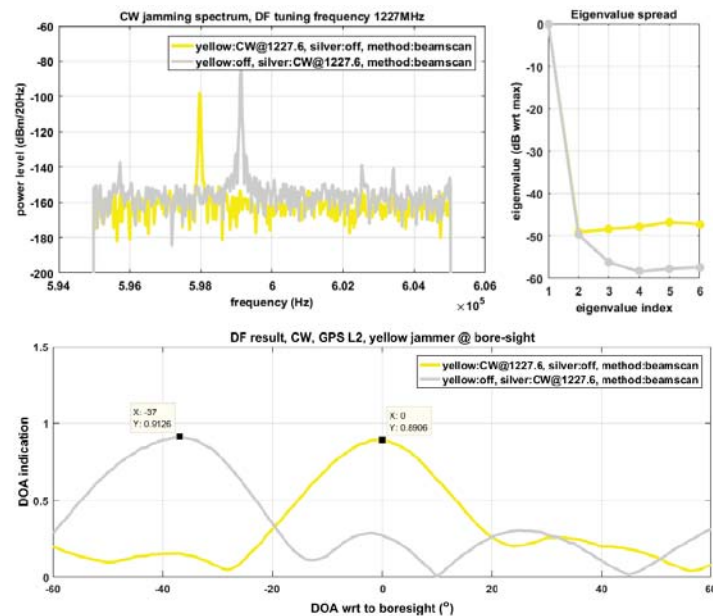


Figure 4. Spectrum, eigenvalue spread and DF result for two single-jammer CW scenarios.

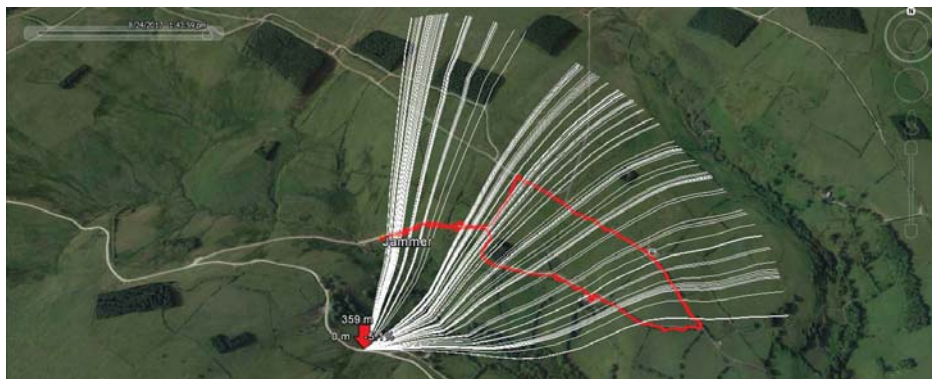


Figure 5. In the red the path of the man-portable jammer. In white, the GANDALF's line of bearings.

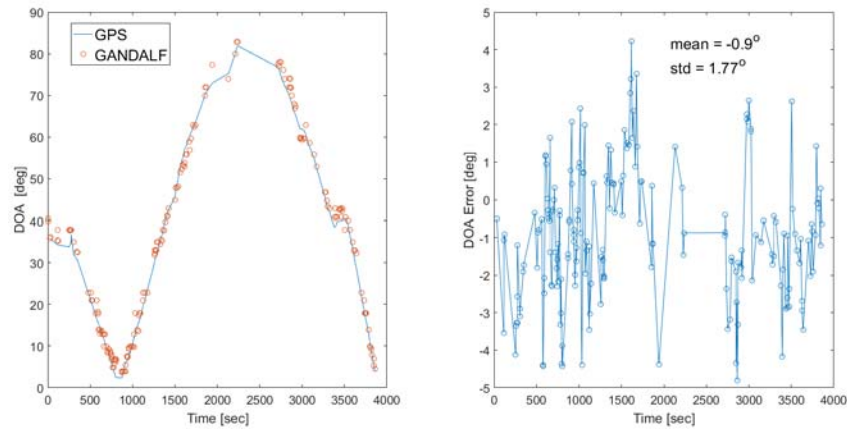


Figure 6. Expected DOAs and measured DOAs (left). Error in DOA (right).

6.3. Spoofing Scenarios

In addition to the jamming scenarios, GPS-L1 spoofing scenarios were investigated too [4]:

- Asynchronous spoofing attack: the spoofer has no (or just a rough) spatial knowledge of the target receiver.
- Meaconing attack: the meaconer receives, delays, and re-broadcasts an entire block of the radio-frequency spectrum containing an ensemble of GPS signals.
- Synchronous spoofing attack: the spoofer first synchronizes to live GPS signals, and then generates the spoofing signal knowing the location of the target receiver with respect to the spoofer.

6.3.1. Asynchronous spoofing attack

In this scenario, the spoofer had to guess both position and velocity of the target receiver. Therefore, to be effective, the spoofer required to change its signal over time and to transmit a power higher than the authentic GPS signal, thus making the attack easy-to-detect.

The main GANDALF's findings, which triggered a spoofing detection, are listed here below:

- a) Some of the satellites, expected not to be visible, were visible instead.
- b) Measured Doppler frequencies, often, not consistent with the expectations.
- c) All detected signals originated from the same direction.
- d) Broad correlation peaks in Doppler domain.

6.3.2. Meaconing attack

This scenario was created by first recording the GPS L1 navigation signals at a point in time during the day and then replaying it, hours later. The meaconer could not arbitrarily manipulate the PVT of the target receiver. Rather, the target receiver was confused and displayed the position and velocity of the meaconer and a time in arrears of true time. In order for the attack to be effective, the meaconer was located in vicinity (within 100 m distance) of the target GNSS receiver. The attack became easy-to-detect because the introduced delay was in the order of several hours rather than minutes.

The main GANDALF's findings, which triggered a spoofing detection, are listed here below:

- a) Some of the satellites, expected not to be visible, were visible instead.
- b) Measured Doppler frequencies, often, not consistent with the expectations.
- c) Too many GPS signals (more than 12) above the acquisition threshold.
- d) All detected signals originated from the same direction.

Figure 7 provides a view of the GANDALF-GNSS toolbox's displaying window, at the time the GPS-L1 C/A spoofing has been detected. In the upper part of the GUI, time domain, frequency domain and distribution of the measured signals are provided. The upper plot on the

right side of the GUI represents the estimated C/N0 for the 32 GPS satellites. These values are obtained by processing the signals measured by the SMA GPS antennas, pointing towards zenith. The bar chart shows in green the signals that are above the acquisition threshold of 34 dB-Hz, in grey the signals from un-healthy satellites and in red the spoofed signals. The bottom plot on the right side of the GUI provides the same information but for the signals measured through the L1 dipole antennas. The corresponding C/N0 values are expected to be lower than those measured by the SMA GPS antennas as the dipoles are pointing towards the battle field. Only in case of low horizon satellites or in case of spoofed signals from the ground, these C/N0 values may be higher than those measured by the SMA GPS antennas. The central panel provides a quick and intuitive view of the current GPS L1 status, with three different flags turning read only when the following events occur: noise jamming, CW (or SS) jamming and spoofing. The content of the text panel is updated every time a new snapshot is analyzed by the tool. The text panel provides useful tables indicating for each acquired channel: the C/N0 level, measured Doppler frequency, measured code offset, status (e.g. healthy, un-healthy or spoofed) and, if spoofed, the reason. Some extra buttons allow the user to change settings, as well as to pause or resume the analysis.



Figure 7: GANDALF-GNSS main GUI. Case of meaconing attack.

6.3.3. Synchronous spoofing attack

In this scenario, the spoofer first synchronized to live GPS signals, and then generated the spoofing signal knowing the 3D pointing vector of the transmit antenna towards the target GNSS receiver antenna. The attack started with a spoofing signal having less power than the authentic signal. The spoofing signal and the authentic GPS signal nearly overlapped. However, due to the difference in power levels, the receiver staid locked to the authentic signal, whereas the spoofing signal was perceived as a multipath effect. The transmit power of the spoofer was gradually increased to allow the fake signal taking over the channels. The GNSS receiver was then dragged off from his original position before the spoofer's power was decreased to conceal the attack. The GANDALF's software noticed the following facts:

- For some satellites, the measured Doppler frequencies was not always consistent with the expectations.
- All acquired GPS signals were originated from the same direction.

By the analysis of the long data recordings, it was possible to retrieve the “virtual” trajectory of the targeted receiver: the spoofer “moved” the receiver in a loop having the shape of the infinity symbol (see Figure 8).



Figure 7. Navigation solution during the synchronous spoofing attack. The fixed receiver was “driven” by the spoofer along a loop.

7. Conclusions

We can conclude that it is possible to make fully phase coherent, multichannel measurements using relatively low cost Software Defined Radio (SDR) technology. That said, phase coherence remains a challenge, especially across separate units. The receive system is not fully phase coherent which must be corrected for by implementing a complicated calibration strategy. Although this is an automated process and requires minimal time, it cannot be done while receiving. These gaps are not an issue if only brief snapshots of data are required, such as for the Direction Finder described in this report. In applications where gaps cannot be tolerated (or only to a limited extent), such as beamforming for GPS reception in a jamming environment, this will be a serious limitation. Other low cost SDR platforms will suffer from the same issues as they are based on similar designs. At the time of writing, end of 2018, SDR models are emerging with Local Oscillator sharing such as the Ettus Twin-RX. These may overcome the above issues. This experiment demonstrated that a small array of only 6 elements can be used to determine direction of arrival with an accuracy of 1° or 2° . The minimal angular separation to resolve two simultaneous jammers is approximately 10° . More accurate calibration of the array manifold will improve this performance but it is difficult to say by how much. Uncontrollable phenomena such as multipath and phase noise, but also repeatability in antenna setup and component stability are likely to be limiting factors. For most GPS jamming related tasks, an accuracy of 1° is more than enough although it depends on the operational goal. The trials have demonstrated that signal analysis (SIGINT) features in the software are extremely useful. The ability to view spectral and temporal characteristics of received RF signals in fine detail provides a trained operator many clues about the signal environment. Thanks to the GANDALF's software, a trained operator can quickly judge what type of jammers (spread spectrum, pulsed, sweep) or spoofers (synchronous, asynchronous or meaconer) are being used and even some waveform parameters. It also allows signals to be resolved in the frequency or time domain, besides only in space.

References

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